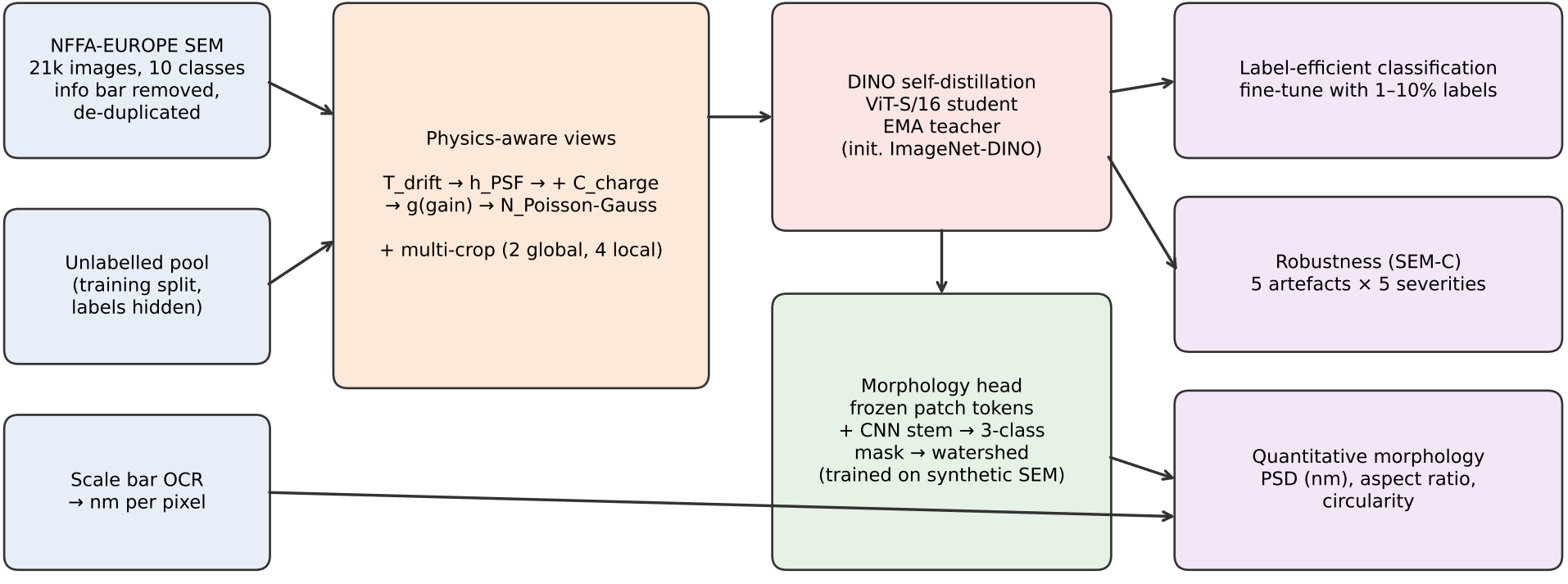
NFFA-EUROPE SEM
21k images, 10 classes
info bar removed,
de-duplicated
Unlabelled pool
(training split,
labels hidden)
Scale bar OCR
→ nm per pixel
Physics-aware views
T_drift → h_PSF → + C_charge
→ g(gain) → N_Poisson-Gauss
+ multi-crop (2 global, 4 local)
DINO self-distillation
ViT-S/16 student
EMA teacher
(init. ImageNet-DINO)
Morphology head
frozen patch tokens
+ CNN stem → 3-class
mask → watershed
(trained on synthetic SEM)
Label-efficient classification
fine-tune with 1–10% labels
Robustness (SEM-C)
5 artefacts × 5 severities
Quantitative morphology
PSD (nm), aspect ratio,
circularity